Oxidative Stress and Skin Aging

Oxidative stress upregulates stress regulatory proteins, including **nuclear factor $\kappa \mathrm{B}$** ($\mathrm{NF} \kappa \mathrm{B}$), a transcription factor that induces the expression of pro-inflammatory cytokines such as **interleukin (IL)-1**, **IL-6**, **vascular endothelial growth factor (VEGF)**, and **tumor necrosis factor (TNF)-α**. These proteins play key roles in immunoregulation and cell survival, stimulate the expression of matrix-degrading metalloproteinases, and are believed to contribute significantly to the aging process.

Oxidative damage also impacts **telomeres**. A proposed hypothesis suggests a shared cellular signaling pathway activated by DNA damage and involving the terminal region of telomeres. The $3^{\prime}$ end of the telomeric strand extends beyond the complementary $5^{\prime}$ strand, forming a single-stranded G-rich overhang. During telomere shortening or repair—particularly under oxidative stress—the normal loop structure (T-loop) at the telomere end is disrupted, exposing the $3^{\prime}$ overhang that is normally sequestered. This exposure of the **TTAGGG** tandem repeat sequence appears to activate **p53**, triggering p53-dependent responses such as **proliferative senescence** and **apoptosis**. Thus, intrinsic skin aging involves progressive oxidative stress and telomere dysfunction due to both replicative shortening and oxidative DNA damage.

Epidermal Changes in Aging

The most consistent histologic change in aged epidermis is **flattening of the dermal-epidermal junction**, with loss of **dermal papillae** and **epidermal rete pegs**. This reduces the surface area for epidermal-dermal interaction, likely impairing communication and nutrient exchange. Aged skin is more prone to dermal-epidermal separation, explaining the increased susceptibility of elderly individuals to skin tearing and superficial abrasions following minor trauma.

There is an **age-associated epidermal thinning of 10% to 50%** between ages 30 and 80. Variability in epidermal thickness and keratinocyte size increases, including in the basal layer. Evidence indicates that **epidermal keratinocytes undergo senescence**, and these senescent cells are more resistant to apoptosis. This resistance may lead to the accumulation of mutations and an increased risk of malignant transformation.

Studies show a **decline in epidermal stem cell populations** in aged skin, as indicated by reduced expression of **CD71 (transferrin receptor)** and **α_6 integrin**, which are established markers of keratinocyte stem cells.

At the ultrastructural level, sun-protected aged skin shows: - Slight widening of inter-keratinocyte spaces - Reduplication of the lamina densa and anchoring fibril complex in the basement membrane zone - Loss of microvillous projections from basal keratinocytes into the dermis

Stratum Corneum and Barrier Function

The average thickness and compaction of the **stratum corneum** remain relatively unchanged with age, although individual **corneocytes become larger**. The skin surface pattern, influenced by underlying papillary dermal architecture, shows a mild loss of regularity with aging.

Age-related changes in percutaneous absorption vary by substance: - **Hydrophilic compounds** (e.g., hydrocortisone, benzoic acid) are less well absorbed in older skin - **Hydrophobic compounds** (e.g., testosterone, estradiol) show similar absorption in young and old skin

Clinically significant is the **delayed recovery of barrier function** in aged skin following stratum corneum damage. This delay is attributed to **slower synthesis and replacement of neutral lipids**, resulting in reduced lipid content in newly formed lamellar bodies. Additionally, **lipid synthesis** and the **activity of enzymes** involved in generating stratum corneum lipids decline with age.